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Original Article

Personality remains: no effect of 3-week social status experience on personality in male fowl

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Individuals often differ in behavior in a consistent way, that is, they show variation in personality. Understanding the processes explaining the emergence and maintenance of this variation is a current major topic in the field of animal behavioral research. Recent theoretical models predict that differences in various "states" can generate individual variation in behavior. Previous studies have mainly focused on endogenous states like metabolic rate or energy reserves, but theory also suggests that states based on social interactions could play important roles in shaping personality. We have earlier demonstrated short-term status-dependent variation in behavior in the domestic fowl (*Gallus gallus domesticus*), but whether such behavioral variation remains also after a longer period of time, is unknown. Therefore, we examine the influence of social status on variation in behavior, using experimental manipulation of social status in pairs of male domestic fowl. We scored males in 3 personality assays (novel arena test, novel object test, and aggression test) before and after 3 weeks in pairs as either dominant or subordinate. We observed individual consistency of behavior despite alteration of social status. We further found no support for social status acting as a state that generates variation in personality over the used time interval: social status had no significant effect on the change in behavioral responses between repeated personality tests. Our results suggest that personality is more important than current social situation for describing individual behavior in stable groups.

Key words: **aggression, behavioral syndrome, chicken, comb size, *Gallus gallus domesticus*, social hierarchy.**

INTRODUCTION

Behavioral responses can vary rapidly and be highly responsive to environmental cues, and theoretically, they should be adapted to current conditions. Therefore, it might be surprising that virtually all species investigated so far across various taxa show among-individual variation and consistency across time and context in behavior, that is, personality (Dall et al. 2004; Sih et al. 2004; Réale et al. 2007). The research field of animal personality has expanded tremendously in the last few decades, and yet the origin and maintenance of consistent variation in personality types is not fully understood. Several adaptive explanations have been proposed, including negative frequency dependent selection, life-history trade-offs, and state-dependency of behavior (Wilson et al. 1994; Dall et al. 2004; Wolf et al. 2007; Biro and Stamps 2008, 2010; Dingemanse and Wolf 2010; Luttbeg and Sih 2010; Wolf and Weissing 2010; Sih et al. 2015). For state-dependence to explain personality, small initial differences in state (e.g., condition, weight, metabolic rate, morphology, behavioral plasticity, or rearing environment) are hypothesized to result in consistent behavioral variation among individuals by positive feedback between state and behavior when costs and benefits of behavior are state-dependent (Wolf and Weissing 2010; Holbrook et al. 2014; Keiser et al. 2015; Sih et al. 2015; Wright et al. 2015). Endogenous differences like metabolic rate, condition, and hormonal state have been suggested as potential states, and lately also exogenous phenotypic traits have been put forward (Bergmüller and Taborsky 2010; Montiglio et al. 2013; Dingemanse and Araya-Ajoy 2015; Sih et al. 2015). In particular, features of the social environment have been emphasized as key states, because of the potential selective forces of social interactions on behavior (Dingemanse and Araya-Ajoy 2015). Repeated social interactions have therefore recently been proposed to be a key component that may generate and maintain individual differences in personality by state-dependency and positive feedback loops between status and behavior (Dall et al. 2012; Briffa et al. 2015; Dingemanse and Araya-Ajoy 2015; Sih et al. 2015). In accordance with the state-dependent personality hypothesis, repeatability is sometimes higher when social groups remain undisturbed, compared to when the social environment is disturbed (Laskowski and Pruitt 2014; Modlmeier et al. 2014). Similarly, the social niche specialization hypothesis predicts that repeated social interactions may generate consistent individual differences not only in social behavior (Aplin et al. 2015), but also in behaviors often studied within

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the context of animal personality (Bolnick et al. 2003; Bergmüller and Taborsky 2010; Montiglio et al. 2013; Saltz et al. 2016). The idea is that adopting different "social niches" within a population decreases competition over resources, which is beneficial to all participants, and that consistency of individual behavior may follow if changing niche is costly (Bergmüller and Taborsky 2010; Montiglio et al. 2013). This hypothesis has rarely been tested empirically, but there are some available data that support the idea. For example, social familiarity increases repeatability of boldness in social spiders (*Stegodyphus mimosarum*, Laskowski and Pruitt 2014; *Stegodyphus dumicola*, Modlmeier et al. 2014). However, there is also empirical data dismissing the idea of social niches as a driver behind consistent individual differences in behavior (3-spined stickleback, *Gasterosteus aculeatus*, Laskowski and Bell 2014).

A well-studied form of repeated social interactions is social dominance relationships, which determine the social structure of group-living species. Dominance relationships are established and maintained through aggressive displays, threats, and attacks by the dominant and submissive acts by the subordinate (e.g., Collias 1943). Dominance increases access to valuable resources (Clutton-Brock and Hutchard 2013). Dominant and subordinate individuals therefore often differ in behaviors related to the access of resources, such as courtship behavior (Desjardin et al. 2012), and foraging tactics (Liker and Barta 2002). It has also been observed that dominant individuals are generally more active, explorative, and bold, which are, behavioral traits that are often used to describe personality (e.g., Korzan et al. 2006; Colletér and Brown 2011). The direction of the causality of the observed relationship between behavior and social status has not been thoroughly investigated. On the one hand, bolder, more explorative, active, or aggressive individuals often have higher chances of obtaining a dominant position (e.g., Dahlbom et al. 2011; Favati, Leimar, and Løvlie 2014). However, social dominance interactions can alter physiology and behavior (Summers and Winberg 2006; Webster and Ward 2011; Carter et al. 2014), indicating that social status may be a state and candidate driver of personality variation (Laskowski and Pruitt 2014; Favati, Leimar, Radesäter, et al. 2014). For example, aggressiveness increases in winners and decreases in losers of social interactions (e.g., Favati et al. 2017) which are known as the "winner–loser effects" in animal contest research (Rutte et al. 2006). Short-term alterations (from hours to a few days) of various other behavioral traits following a dominance interaction have also been observed, such as boldness, exploration, and vigilance (Courtene-Jones and Briffa 2014; Favati, Leimar, Radesäter, et al. 2014; Rudin et al. 2017). However, it is unknown if having a certain social status for a longer period of time may result in the observed differences between dominant and subordinate individuals. Further empirical testing is thus needed to evaluate the time aspect in the role for social interactions as a stabilizing state.

The domestic fowl, *Gallus gallus domesticus*, is a group living species where hierarchical dominance relations determine access to resources, in particular male access to mating partners (Collias and Collias 1996). Males with larger combs, a fleshy red ornament on the head, are more often found in a dominant position (e.g., Johnsen et al. 2001). Comb size sometimes predicts social dominance status, which indicates that the comb reflects dominance ability (Zuk and Johnsen 2000). Comb size also increases when a male holds a dominant position (Zuk and Johnsen 2000; Parker et al. 2002; Cornwallis and Birkhead 2008). This indicates that comb size is at least partially a socially mediated ornament and a signal of current social status. Similar interplay has been observed between dominance and behavior. When controlling for morphological differences (comb size and body size), aggressiveness and explorative behavior can to some extent predict dominance in male domestic fowl (Favati, Leimar, Radesäter, et al. 2014), and these behaviors can change in magnitude following a change in social status (Favati, Leimar, and Løvlie 2014; Favati et al. 2017). However, it is unknown whether such behavioral differences between dominant and subordinate individuals remain in the longer term, after more transient behavioral changes due to a recent change in status (e.g., winner–loser effects, Rutte et al. 2006) have worn off.

We here use an experimental approach to study the effect of social status on individual behavior in male domestic fowl, using an extended time scale (3 weeks), compared to earlier short-term dominance interactions (from hours to days) shown to affect behavior in several species, including the domestic fowl. If social status is a state that can create lasting behavioral disparity in the form of different personality types, we expect dominant and subordinate males to differ in behavioral responses after the treatment period. We further expect combs to grow in males that become dominant (and possibly to shrink in males that become subordinate).

MATERIALS AND METHODS

Animals and housing

The study was conducted during the breeding season (April to July, 2014) when social status is particularly important for males, at Tovetorp research station, Stockholm University, Sweden. Forty-four males (1–6 years old, i.e., all birds used were sexually mature) of the breed Swedish bantam ("Gammalsvensk dvärghöna" in Swedish), were used. Prior to the study, 38 of the males were housed in 6 mixed-sex (sex ratio ~1:1), mixed-age (1–9 years old) groups (4–16 individuals per group), and 6 of the males (1 year old) were recently separated from females and housed in a group with only males (due to that the females took part of another experiment). Birds were housed in roofed outdoor aviaries (~4 × 6 m) that were floored with sand and fitted with nest boxes, perches and branches with fresh leaves. Water and food were available ad libitum. Birds in different aviaries could hear, but not see each other. During the experiment birds were housed in smaller aviaries (2–3 × 3 m) otherwise similar to the regular aviaries. All behavioral observations took place during the birds' active hours (7:30 am to 11:00 am and 2:00 pm to 6:00 pm, Løvlie and Pizzari 2007). The study fulfilled the ethical requirements in Sweden (Linköping ethical committee, permit no. 114 - 12).

Social status treatment

There are 2 principal ways to experimentally produce dominant and subordinate individuals. The "randomized" method is to pair focal individuals with individuals that are either likely to become dominant (e.g., a larger resident) or subordinate (e.g., a smaller intruder, see discussion in Hsu et al. 2006). The drawback of this method is that winners and losers do not receive the same experimental treatment, making it difficult to directly compare the treatment groups. Alternatively, as adopted in this study, 2 individuals can be paired to settle the dominance relationship, resulting in "self-selected" dominance roles where both contestants experience the same social interaction, except the outcome (Beaugrand and Goulet 2000). Such winners and losers are likely to differ in traits important for dominance. We therefore undertook several precautions to increase the chances that the winner and loser were

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randomly selected within each pair: males were closely matched for phenotypical traits that otherwise may interfere with the outcome (comb size and body size), and we only used males that previously were intermediately ranked (see further details below). Moreover, we investigated statistically, post-pairing that none of the personality traits in focus had a significant impact on the outcome of obtained status in a pair.

Twenty-two pairs of males were assembled to create dominance relationships and thereafter housed together for 3 weeks, which should be long enough to affect comb size and behavior (Cornwallis and Birkhead 2008) and to exclude any winner–loser effects (Rutte et al. 2006). Males were matched within pairs for comb size (comb length measured to the nearest mm, maximum 10% difference within a pair), body weight (g, maximum 20% difference), and age (0–2 years difference, except one pair that had a 4-year difference). In order to reduce any potential effects of previous social status, we ensured that males within a pair had not been housed together for at least 3 weeks prior to the experiment (Chase 1982; Cornwallis and Birkhead 2008; Favati, Leimar, Radesäter, et al. 2014), and only males with a previous intermediate social position were used. The top and bottom ranked males were identified in each pre-experimental group ($n = 7$) by observation of dominance interactions among males, using 3–5 avoidances to classify a male as subordinate to another. However, due to a lack of matching intermediate ranking males, 2 of the top ranked males were used as pair mates in the experimental pairs, but were not included in any analysis. Each pair was initially observed until 5 successive avoidances of the same male had been observed, which was used to classify him as subordinate and the other male as dominant (sensu Favati, Leimar, Radesäter, et al. 2014). Weekly observations ascertained that the males kept their social positions throughout the study. Each pair of males was accompanied by 2 females for a natural group composition (Collias and Collias 1996).

Comb size (estimated by comb length) was measured before and after the 3-week status treatment in order to investigate the effect of social status on comb size.

Personality assays

In order to investigate the effect of social status on individual behavior, 3 personality assays (novel arena test, novel object test, and aggression test) were performed twice: the first time 1–3 days before pair formation, and the second time after 3 weeks of having remained in these pairs (Figure 1). The species has earlier shown significant repeatability of individual behavioral responses in these assays (Favati, Leimar, Radesäter et al. 2014; Favati et al. 2016; Favati et al. 2017; Zidar, Sorato, et al. 2017), and we therefore only performed each assay one time before and one time after the social treatment period. At each of the 2 test occasions,

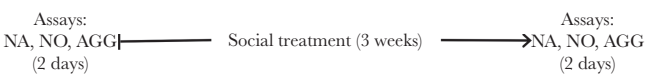

Figure 1
Experimental design. Male domestic fowl were housed in pairs for 3 weeks, which means males were either dominant or subordinate during this period of social treatment. Three personality assays ("NA", novel arena test; "NO", novel object test; "AGG", aggression test) were performed twice: one time before, and one time after the social treatment period. The 3 assays were performed during the course of 2 days, with one test the first day and 2 the second, or vice versa, for each male, in random order.

each male was singly subjected to one or 2 personality assays per day during 2 successive days. The novel arena test was performed in the morning (7:30 am to 11:00 am) on one of the 2 days, and the novel object test was performed in the afternoon (2:00 pm to 6:00 pm) on one of the days, and the aggression test was performed in the afternoon on the other day. The order of assays was randomized among individuals. For the novel arena test, the focal male was gently caught and transported to the test arena. The two other tests were performed in the home aviary, while the other birds in a group were temporarily herded to an empty adjacent aviary.

Novel arena test

A novel arena test was performed for 15 min in an outdoor fenced arena that was oval shaped (11.5 × 10 m) and divided into 32 roughly even sized areas, situated in a deciduous forest (for a detailed description, see Favati, Leimar, and Løvlie 2014). Behaviors observed were "vigilance" (proportion of observations where the male had the head over shoulder height, observed every 30 s), "activity" (number of area transitions), "exploration" (number of unique areas visited), and "crowing" (number of territorial crows uttered).

Novel object test

A novel object test was conducted to estimate boldness. The novel objects used were a multicolored "hairy" dog toy and a multicolored rope knot, both approximately 3 × 15 cm. One of the objects was used during the first novel object test (before the social treatment), and the other the second time (after the social treatment), in random order. The 2 objects were similar in size and color, and elicited equivalent responses of avoidance, and we therefore consider the 2 tests as a repetition of the novel object test as such. The males were trained once to eat mealworms (highly desired food) from a food plate at one end of their home aviary. Thereafter, the male was gently herded to the opposite side of it's home aviary, and the observer placed a mealworm on the plate and a novel object 2 cm in front of the plate. Thereafter the observer ensured that the male was approximately 2 m away from the object, backed off and started a stopwatch. Latency to approach the food plate (and eat the mealworm) was noted, and used as an inverted estimate of boldness (shorter latency represents a bolder individual, and a longer latency represents a shyer one). Males that did not approach the plate within 10 min were assigned a maximum score of 600 s. Around half of the males received maximum values, and we therefore converted the variable to a binary response where males that reached the plate within the time limit were classified as 1, and those who did not were classified as 0.

Aggression test

Aggression was estimated as the agonistic reaction of the focal male towards a conspecific nonfocal male intruder that was restrained in the hands of the observer. The intruder was presented for 1 min, or until the focal male initiated an attack, whichever came first (for further details, see Favati et al. 2017). Aggressiveness (aggressive body posture, displays and approaches/attacks) was scored along a scale from 0 to 6 where 0 was the least aggressive response and 6 the most aggressive (sensu Favati et al. 2017, Table 1). Nonfocal intruder males ($n = 12$) had not been housed together with experimental males for at least 3 weeks, and had equal or maximally 10% smaller combs compared to the focal male.

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Table 1

Scoring of aggression in male domestic fowl in response to presentation of a manually restrained male conspecific

Score	Description
0	Walks away from the other male
1	Neutral body posture, no avoidance or attraction to the other male
2	Aggressive posture after > 30 s
3	Aggressive posture within 30 s
4	Aggressive posture + aggressive ground pecking within 30 s and/or approach opponent < 50 cm within 60 s
5	Aggressive posture + approach opponent < 50 cm within 30 s
6	Approach opponent < 50 cm within 10 s

The aggression scores range from 0 to 6 (6 being highest). An aggressive posture consisted of one or more of the following: tilted and/or crouched posture, lowered wing, raised hackles (Collias 1943).

Statistical analysis

Paired Wilcoxon tests were used to confirm successful matching of males within pairs for comb size, body weight, and age, and to investigate whether paired males differed in behavior before the treatment started.

To estimate the effects of social status treatment and personality on behavioral variation, linear models were fitted to each of the behavioral response variables. Aggression and exploration were approximately normally distributed and were fitted to linear models with a Gaussian distribution of residuals. Activity and crowing were counts that were fitted with generalized linear models with Poisson distributed response. Vigilance (proportion of time being vigilant vs. nonvigilant) and boldness (binomial variable, approached vs. did not approach) were analyzed using logistic regression models. For all models, the response in the second round of personality assays (after treatment) was used as the dependent variable, and social status and response in the first round of personality assays (before treatment) were entered as explanatory variables. To investigate if males altered their behavior according to their new status following the 3-week status treatment, the interaction term between status and previous behavioral response was included in all models.

To investigate the relationship among behavioral responses, Spearman rank correlation using mean values of the 2 trials (before and after treatment) were performed. *P*-values were adjusted using the false discovery rate procedure for multiple testing (Benjamini and Hochberg 1995).

The effect of social status on comb length was analyzed by linear mixed model with comb length after the treatment period as response variable, and social status and comb length before the treatment period as fixed effects, and "pair" as a random factor. The interaction between the 2 fixed effects was significant which means that either small combed or large combed males changed more. Therefore, further investigation of the difference in comb length between dominant and subordinate males was performed by additional linear regressions with the intercept set to 60 (which represents the size of a small comb) and 90 (which represents the size of a large comb), respectively, to determine in which comb size region males differed.

All statistics analyses were conducted using the statistical software R v. 3.3.1 (R Core team 2016) and packages lme4, MASS and psych.

RESULTS

Pairs of males were successfully matched for comb size ($mean_{dom}$ = 76.6 mm ± 2.5 SE, $mean_{sub}$ = 76.9 mm ± 2.4 SE, mean difference: 0.04 mm ± 0.88 SE, V = 551, P = 0.49), body weight ($mean_{dom}$ = 1165 g ± 28 SE, $mean_{sub}$ = 1193 g ± 31 SE, mean difference: 17 g ± 27 SE, V = 432, P = 0.47), and age ($mean_{dom}$ = 3.6 years ± 0.5 SE, $mean_{sub}$ = 3.6 years ± 0.4 SE, mean difference: 0.01 years ± 0.23 SE, V = 53, P = 1).

Effects of social status on behavior

There were no significant differences in behavioral responses between dominant and subdominant males before the males were paired (all $P > 0.10$). After the 3-week status treatment, dominant males did not differ from subordinate males in aggression, exploration, crowing or boldness, but dominant males were less vigilant and more active than subordinate males (Figure 2, Table 2). However, the interaction term between rank and previous behavioral response was not significant in any case, including vigilance and activity (Table 2). In other words, social rank did not have a significant effect on the change in behavior between the 2 observations.

Consistency of behavior

Consistency is represented by a significant effect of the first time a behavior was measured on the second measure of the same behavior in linear models (Table 2). Male domestic fowl showed significant among-individual consistency in aggression, activity, boldness, and crowing (Figure 2, Table 2). Vigilance and exploration showed positive but nonsignificant correlations across time and treatment (Figure 2, Table 2).

Correlations among traits

Activity and exploration were strongly positively correlated, most probably due to that the number of area transitions (activity) was low and thus in several cases were translated into being the same as the number of *unique* areas visited (exploration). No other behavioral responses were significantly correlated (Table 3).

Effects of social status on comb size

There was a statistically significant interaction between social status and comb size before treatment (F = 6.8, $P < 0.009$; Figure 3). A closer examination of this interaction revealed that the combs of males in a dominant position were slightly larger after the treatment than the comb of males that had been in a subordinate position, but this effect was only observed for males with initially small combs (intercept set at 60 mm, difference −2.55 mm; intercept set at 90 mm, effect size 0.44 mm). Overall, comb size was among-individually consistent across the treatment period (r = 0.98, $P < 0.0001$), and differences in comb size between dominant and subordinate males were not larger after compared to before treatment (difference dominant - subordinate before treatment ranged from −9.2 to 5.3 mm; and after from −7.2 to 6.3 mm).

DISCUSSION

We here show that being dominant or subordinate for 3 weeks during the breeding season had no notable effect on the behavioral responses of males in 3 different personality assays. This result indicates that variation in behavior is not a long-term consequence

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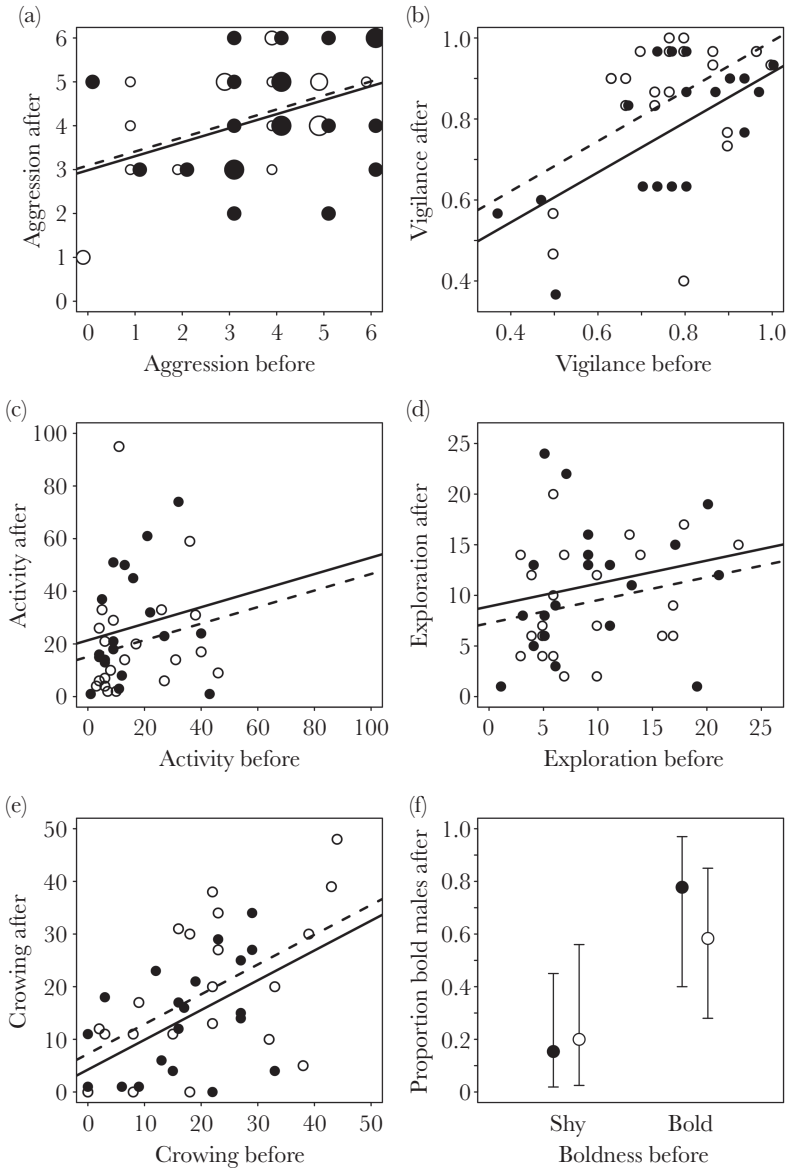

Figure 2

Behavior of male domestic fowl in repeated personality assays. Males were assayed before and after a 3-week stay in male-pairs causing one male to be dominant (filled dots, solid line) and the other subordinate (empty dots, dashed line). All behavioral responses were significantly consistent (i.e., positive slopes), except activity where a positive trend was observed. There was no significant difference between dominant and subordinate males in (a) aggressiveness (aggression scores, radius of dots is proportional to the number of males with that aggression score). In the novel arena test, dominant and subordinate males did not differ significantly in (b) time spent being vigilant, (c) activity (number of subareas transitions), (d) exploration (number of subareas visited), or (e) crowing (number of territorial crows). There was neither a significant difference in (f) boldness in the novel object test between dominant and subordinate males. Error bars are 95% CI. Plots are of raw data.

of variation in social status, in other words that personality is not dependent of social situation in stable groups. We further confirm, as previously observed, that male domestic fowl were consistent in their behavioral responses in these personality assays. This is in line with earlier results (Favati, Leimar, Radesäter, et al. 2014), and indicates that male fowl exhibit robust personality types.

Personality traits and social status often correlate, and the general trend is that bolder, more explorative or aggressive individuals are observed in dominant positions (e.g., Kralj-Fiser et al. 2010; McGhee and Travis 2010; David et al. 2011). One explanation to this relationship is that personality at least partly determines the chances of obtaining a dominant position (e.g., Verbeek et al. 1996; Dahlbom et al. 2011). In contrast to an earlier study of the same population of male domestic fowl (Favati, Leimar, and Løvlie 2014), social status was here not predicted by personality type. However, the current study was not specifically designed to test this question. In fact, we deliberately only used males of initially intermediate social status to avoid impact of earlier status, thereby excluding both extremes of socially dominant and socially subordinate individuals. The relationship previously observed between personality and social status may depend on larger initial variation in personality, such that individuals with very high or very low aggressiveness end up at the opposite end of the dominance hierarchy. For example, in zebra fish (*Danio Rerio*), double wins/losses were used

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Table 2

Effect of status and individual on behavioral responses of dominant and subordinate domestic male fowl after 3 weeks in stable pairs

Response	Mean (± SE) Dominant	Mean (± SE) Subordinate	Effect	Estimate (± SE)	P
Aggression	4.2 ± 0.29	4.18 ± 0.28	Status	0.88	0.78
			Aggression time 1	0.17	<0.01
			Status:Aggression time 1	0.27	0.23
Vigilance	0.77 ± 0.04	0.85 ± 0.04	Status	−0.22	<0.001
			Vigilance time 1	0.12	<0.001
			Status:Vigilance time 1	−0.01	0.70
Activity	2.92 ± 0.22	2.64 ± 0.20	Status	0.24	<0.001
			Activity	0.04	<0.001
			Status:Activity	−0.00	0.65
Exploration	3.14 ± 0.23	2.94 ± 0.19	Status	2.15	0.37
			Exploration time 1	0.20	0.16
			Status:Exploration time 1	0.05	0.86
Crowing	14.0 ± 2.2	18.5 ± 3.0	Status	−0.16	0.08
			Crowing time 1	0.03	<0.001
			Status:Crowing time 1	−0.00	0.90
Boldness	0.40 ± 0.11	0.41 ± 0.11	Status	−0.11	0.56
			Boldness time 1	0.79	<0.001
			Status:Boldness time 1	−0.41	0.11

The outcome of models exploring the effect of social status and personality on behavior after 3 weeks of being either dominant or subordinate (n_{dom} = 20, n_{sub} = 22). The effect of "status" is the dominant-subordinate difference and the interaction "status:behavioral response" is the deviation from additivity for the before-after difference for subordinates. The interaction term status:behavioral response before treatment was in no case significant, that is, social status did not have a significant effect on the change in behavior between observations. The behavioral response before treatment ("time 1") had a significant impact on the response after treatment (the dependent variable) for all responses except for exploration, where there was a positive but nonsignificant correlation.

Table 3

Correlation matrix of behavioral responses in 3 personality assays of male domestic fowl

	Aggression	Vigilance	Activity	Exploration	Crowing	Boldness
Aggression	-	−0.29	−0.10	−0.16	0.17	0.07
Vigilance	0.06	-	0.28	0.26	−0.26	−0.06
Activity	0.54	0.07	-	0.89	0.00	−0.32
Exploration	0.32	0.10	**<0.0001**	-	−0.02	−0.29
Crowing	0.28	0.09	0.99	0.89	-	−0.11
Boldness	0.64	0.71	0.04	0.06	0.50	-

Mean response of 2 times (before and after treatment) was used to calculate Spearman's *r* (above the diagonal) and p-values (below the diagonal) of the correlations among responses. *P*-value in bold was significant also after Benjamini and Hoch's correction for multiple testing.

to create a selection of individuals that were particularly prone to win or lose, respectively, before testing whether their personality type could predict the status later obtained (Dahlbom et al. 2011). If more extreme individuals are the main drivers of the general correlation of personality and dominance, this could explain why we here did not detect such a relationship when using only males of intermediate rank.

Understanding the causes of individual variation in personality is a major current question in the study of animal behavior. It has been suggested that when individuals differ in various states that affect costs and benefits of behavior, this can in turn cause variation in behavior (Dall et al. 2004; Wolf et al. 2007; Biro and Stamps 2010; Wolf and Weissing 2010; Sih et al. 2015). Positive feedback loops between state and behavior are suggested to stabilize observed variability at least as long as the state differences remains, but potentially longer due to carry-over effects (e.g., learning, effects on cost-benefit balance, Briffa et al. 2015). The state-dependent behavior hypothesis can be applied to the hierarchical social structure often found in group-living animals resulting in that social status is a potential state that is associated with both behavioral variation (Courtene-Jones and Briffa 2014; Favati, Leimar, and Løvlie 2014) and fitness outcomes (Smith and Blumstein 2008). The social environment may thus have an impact on variation and consistency of individual behavior (Carter et al. 2014; Favati, Leimar, Radesäter et al. 2014; Laskowski and Pruitt 2014; Modlmeier et al. 2014; Rudin et al. 2017). We have earlier shown that male domestic fowl differ in vigilance, activity, and exploration in a similar setup where behavior was measured in a novel arena test shortly (2 days) after males had been experimentally manipulated to become either dominant or subordinate (Favati, Leimar, Radesäter et al. 2014). In the current study, we allowed the social relationship to be maintained for a longer period of time, and consequently did not observe any significant differences in behavior between males of different status. This is in line with a long-term study of Seychelles warblers (*Acrocephalus Sechellensis*, Edwards et al. 2016). Similarly, aggressiveness is influenced in the short term by an experience of winning or losing contests in the fowl and other species (Hsu et al. 2006; Favati et al. 2017), but here we observed no statistically significant differences in aggressiveness between males of different status when the social relationship had been established for 3 weeks. When losing a social interaction over dominance, short-term suppression of aggressiveness and possibly also activity is appropriate to avoid further attacks

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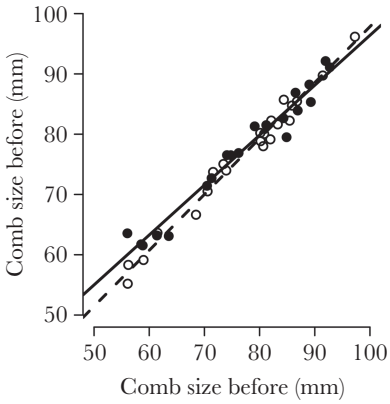

Figure 3
Comb size (estimated by comb length) of male domestic fowl before and after having either a dominant (filled dots, solid line), or a subordinate (empty dots, dotted line) social status for 3 weeks. Among males with initially smaller combs, dominant males increased more in comb size compared to subordinate males. This status effect was not observed in males with initially larger combs.

from the winner. Changes in aggressiveness and activity, so called "winner-loser effects," generally fade away over a few days or even hours after the contestants have been separated (Hsu et al. 2006; Trannoy et al. 2016). Few studies have investigated whether winner–loser effects remain if the social situation remains stable, and a time aspect may be important for the impact of social dominance on behavior. In other words, behavioral adjustments to a new social situation may be strongest shortly after group reorganization and become more relaxed when some time have passed and the dominance relationships are stable.

In addition to the likely influence of time in a social position, behavioral differences between dominant and subordinate animals may be affected by the social setting during behavioral observation (Webster et al. 2007). Vigilance in male domestic fowl has earlier been shown to depend on social status in stable groups (1–4 weeks of undisturbed groups, Cornwallis and Birkhead 2008). Their observations were performed in a social context with familiar males and females present, while we removed the male from his group and observed him in a novel arena. The presence of competitors and sexual partners most likely brings an increased need for dominant males to keep an eye on rivals, while subordinate males spend more time eating, which could explain why Cornwallis and Birkhead (2008) observed a difference in vigilance, while we did not. Further, personality may override dominance also when observed in a social context: activity, but not social dominance, predicted leadership in foraging producer-scrounging games in zebra finches (*Taenopygia guttata*, Beauchamp 2000). From a neuroendocrine perspective, serotonin has an inhibitory effect on aggressiveness in subordinate individuals (Summers and Winberg 2006). However, serotonergic state changes quickly (within seconds to minutes) in encounter with a new opponent, so that also previous subordinate individuals become ready to fight (Summers and Winberg 2006). The lack of difference in aggression between dominant and subordinate males may have been due to that the pair mate was removed before assaying aggression, thereby removing the suppression of the dominant male (Mench and Ottinger 1991) and providing time to change serotonergic state.

Boldness often correlates with other "proactive" personality traits such as activity, exploration and aggression (Koolhaas et al. 1999), and relates to social dominance, both by postfighting changes in boldness (Rudin and Briffa 2012) and predicting the outcome of contests (e.g., Dahlbom et al. 2011). Behaviors related to risk-assessment and recovery from stress are hypothesized to be particularly closely related to social dominance (Briffa et al. 2015). Yet, we observed that boldness was unaffected by changes in social status, in accordance with earlier studies (Favati, Leimar, Radesäter, et al. 2014; Rudin et al. 2017). Further, our results do not support the presence of a behavioral syndrome among these behavioral responses, confirming earlier results in the species (Favati, Leimar, Radesäter, et al. 2014; Favati et al. 2016; Zidar, Balogh, et al. 2017), suggesting that these traits have different underlying mechanisms.

Previous studies on the fowl have shown status-dependent suppression of comb size (Zuk and Johnsen 2000; Parker et al. 2002; Cornwallis and Birkhead 2008). In the current study, comb size of subordinate males did not decrease. This can probably be because we used males with previous intermediate rank that already had suppressed comb sizes when the study started (Parker et al. 2002). However, dominant males increased in comb size, although the effect was very small: the increase in comb size was smaller than the initial difference between dominant and subordinate males. This increase was further dependent on initial comb size: males with initially small combs increased more in comb size compared to larger combed dominant males. The differential comb growth was possibly due to that small-combed males were allowed to grow closer to their physiological potential, while a "ceiling effect" may have prevented the large-combed males from an equivalent growth (Parker et al. 2002). However, the increased growth in dominant males indicates that males indeed did experience the change in social status, although behavior in personality tests did not exhibit a corresponding change.

In summary, personality appears a stronger determinant of individual behavior compared to current social position in stable pairs of male domestic fowl. While behavior was affected over the short-term in an earlier study, similar effects were not found here, where dominance relationships were stable for several weeks. The effect of social status on behavior thus appears to be context dependent, possibly reflecting underlying physiological time-restricted winner–loser effects. The exact circumstances under which social interactions promote variation in personality types remain to be clarified, before drawing further conclusions of the generality and causality of the relationship between social status and personality variation.

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Data accessibility: Analyses reported in this article can be reproduced using the data provided by Favati, Udén et al (2017).

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